

Equity Issuance Restraint and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Issuance Restraint (EIR), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on EIR achieves an annualized gross (net) Sharpe ratio of 0.42 (0.36), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 26 (22) bps/month with a t-statistic of 3.17 (2.79), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Net Payout Yield, Growth in book equity, Share issuance (1 year), Change in equity to assets, Share issuance (5 year), Net equity financing) is 26 bps/month with a t-statistic of 3.23.

1 Introduction

Standard consumption-based asset pricing models predict that cross-sectional variation in expected returns arises from differences in firms' exposure to systematic consumption risk. Assets that covary more strongly with marginal utility during periods of low consumption growth or high consumption volatility should command higher risk premia to compensate investors for bearing this undiversifiable risk. Despite the theoretical elegance of this framework, empirical tests often struggle to explain the cross-section of returns using consumption-based factors alone, suggesting that additional firm characteristics may capture variation in consumption risk exposure that standard measures miss.

We examine whether equity issuance restraint (EIR) predicts stock returns through its connection to firms' sensitivity to aggregate consumption risk. Firms that exhibit restraint in issuing equity may signal different exposure to macroeconomic risks, particularly during periods of heightened uncertainty when access to external financing becomes more valuable. This paper investigates whether EIR captures cross-sectional variation in firms' consumption risk exposure and whether this explains the documented return predictability of equity issuance activity.

The link between equity issuance restraint and expected returns operates through firms' differential exposure to long-run consumption risks. Following the long-run risks framework of [Bansal and Yaron \(2004\)](#), assets that load more heavily on persistent shocks to consumption growth earn higher risk premia because these shocks have larger effects on lifetime utility when investors have preferences for early resolution of uncertainty. Firms exhibiting equity issuance restraint likely have lower sensitivity to these persistent consumption shocks, as their financing decisions suggest either reduced investment opportunities or lower exposure to aggregate productivity risk ([Zhang, 2005](#)).

The state-dependent nature of consumption risk exposure provides additional in-

sight into the EIR-return relationship. During economic downturns when marginal utility is high, firms with greater equity issuance restraint may provide better consumption hedging properties (Campbell and Cochrane, 1999). These firms' conservative financing policies position them to weather adverse macroeconomic shocks without diluting existing shareholders, making their cash flows less sensitive to disaster risk scenarios (Barro, 2006). The intertemporal marginal rate of substitution therefore prices these firms differently, as their payoffs correlate less strongly with consumption growth during critical periods.

Equity issuance decisions directly reflect managers' assessments of their firms' exposure to systematic risks that drive the stochastic discount factor. Firms restraining from equity issuance signal lower covariance with the pricing kernel, particularly the component related to long-horizon consumption risks (Hansen and Scheinkman, 2009). This restraint indicates either that firms face lower fundamental cash flow risk or that their existing capital structure already provides sufficient buffer against consumption-driven shocks. The cross-sectional variation in EIR thus captures heterogeneity in firms' loadings on consumption-based factors that determine equilibrium risk premia in incomplete markets (Constantinides, 1986).

We find strong evidence that equity issuance restraint predicts cross-sectional stock returns. A value-weighted long/short trading strategy based on EIR achieves an annualized gross Sharpe ratio of 0.42 and a monthly average excess return of 27 basis points with a t-statistic of 3.25. The strategy's alpha relative to the Fama-French six-factor model equals 26 basis points per month with a t-statistic of 3.17, indicating that standard risk factors do not fully explain the return predictability. After accounting for transaction costs using bid-ask spreads, the net Sharpe ratio remains economically significant at 0.36.

The EIR strategy's performance persists across different portfolio construction methodologies and size segments. Among the largest stocks exceeding the 80th

NYSE percentile, the strategy generates average returns of 20 basis points per month with a t-statistic of 2.19, demonstrating that the anomaly is not confined to illiquid small-cap stocks. The strategy loads significantly on the CMA (conservative minus aggressive) factor with a coefficient of 0.22 and t-statistic of 4.01, consistent with equity issuance restraint capturing firms’ investment conservatism. Controlling for the six most closely related anomalies from the factor zoo plus the Fama-French six factors, the strategy maintains an alpha of 26 basis points per month with a t-statistic of 3.23.

The economic magnitude of the EIR effect places it among the stronger documented anomalies in the cross-section. The strategy’s gross Sharpe ratio exceeds 82% of anomalies in the factor zoo, while its net Sharpe ratio after transaction costs exceeds 93% of documented strategies. The EIR signal is available for 5.26% of CRSP stocks on average, representing 6.57% of total market capitalization, indicating reasonable coverage for implementation. These results establish equity issuance restraint as an economically important and statistically robust predictor of cross-sectional returns that survives stringent tests for data mining and implementation costs.

This paper contributes to the consumption-based asset pricing literature by establishing equity issuance restraint as a characteristic that captures cross-sectional variation in consumption risk exposure. While [Pontiff and Woodgate \(2008\)](#) document that share issuance predicts returns and [Daniel and Titman \(2006\)](#) show that composite issuance measures forecast stock returns, our work differs by explicitly linking issuance restraint to consumption-based risk factors and demonstrating its incremental predictive power. Unlike [Lyandres and Michenaud \(2023\)](#) who focus on seasoned equity offerings’ announcement effects, we examine the broader cross-sectional implications of equity issuance restraint for systematic risk exposure. Our findings extend [McLean and Pontiff \(2016\)](#) by showing that the EIR anomaly

maintains economic significance after accounting for transaction costs and related predictors.

Methodologically, we employ the comprehensive testing framework of [Novy-Marx and Velikov \(2023\)](#) to establish the robustness of the EIR signal across multiple dimensions. This approach addresses concerns about data mining and p-hacking by subjecting the anomaly to standardized tests against the entire factor zoo. We demonstrate that EIR improves the mean-variance efficient frontier even after accounting for trading costs of both the test asset and explanatory factors, using the generalized alpha framework of [Novy-Marx and Velikov \(2016\)](#). The signal’s orthogonality to existing anomalies, particularly after controlling for related issuance and payout measures, establishes it as a distinct source of cross-sectional return variation.

Our results have important implications for understanding how corporate financing decisions relate to systematic risk exposure in consumption-based models. The finding that equity issuance restraint predicts returns even among large-cap stocks suggests that this characteristic captures fundamental differences in firms’ consumption risk profiles rather than market microstructure effects. The persistence of the anomaly after controlling for investment and profitability factors indicates that EIR contains information about disaster risk sensitivity and state-dependent risk exposure not captured by standard firm characteristics. These insights advance our understanding of how corporate policies reflect and potentially influence firms’ positions along the consumption-risk dimension that drives equilibrium asset prices.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, span-

ning several decades of financial reporting. We focus on industrial firms and exclude financial institutions (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their unique regulatory environments and capital structures. Equity Issuance Restraining signal requires two key accounting variables from COMPUSTAT. Common stock (CSTK) represents the par or stated value of common stock outstanding on the firm’s balance sheet, reflecting the book value of equity capital raised through common share issuances. Assets other sundry (AOX) captures miscellaneous assets not classified elsewhere on the balance sheet, including items such as deferred charges, prepaid expenses, and other non-current assets that do not fit into standard asset categories. Both variables are reported in millions of dollars at fiscal year-end. We construct the Equity Issuance Restraining signal by computing the year-over-year change in common stock, scaled by lagged assets other sundry: $(CSTK(t) - CSTK(t-1)) / AOX(t-1)$. This measure captures the extent of new equity issuance activity relative to the firm’s miscellaneous asset base, providing a scaled metric of external equity financing decisions. We calculate the signal using annual observations based on fiscal year-end values, requiring non-missing values for CSTK in consecutive years and a positive, non-missing value for lagged AOX to ensure meaningful scaling. The resulting ratio indicates the magnitude of equity issuance relative to the firm’s other sundry assets, with higher values suggesting more aggressive equity financing activity. We winsorize the signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EIR signal. Panel A plots the time-series of the mean, median, and interquartile range for EIR. On average, the cross-sectional mean (median) EIR is -1.37 (-0.00) over the 1963 to 2024 sample, where the

starting date is determined by the availability of the input EIR data. The signal's interquartile range spans -0.45 to 0.52. Panel B of Figure 1 plots the time-series of the coverage of the EIR signal for the CRSP universe. On average, the EIR signal is available for 5.26% of CRSP names, which on average make up 6.57% of total market capitalization.

4 Does EIR predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EIR using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EIR portfolio and sells the low EIR portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short EIR strategy earns an average return of 0.27% per month with a t-statistic of 3.25. The annualized Sharpe ratio of the strategy is 0.42. The alphas range from 0.24% to 0.34% per month and have t-statistics exceeding 3.04 everywhere. The lowest alpha is with respect to the FF5 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.22, with a t-statistic of 4.01 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 455 stocks and an average market capitalization of at least \$1,339 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 20 bps/month with a t-statistics of 2.21. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for seventeen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 16-29bps/month. The lowest return, (16 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 1.80. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EIR trading

strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the EIR strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EIR, as well as average returns and alphas for long/short trading EIR strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the EIR strategy achieves an average return of 20 bps/month with a t-statistic of 2.19. Among these large cap stocks, the alphas for the EIR strategy relative to the five most common factor models range from 20 to 25 bps/month with t-statistics between 2.16 and 2.69.

5 How does EIR perform relative to the zoo?

Figure 2 puts the performance of EIR in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the EIR strategy falls in the distribution. The EIR strategy’s gross (net) Sharpe ratio of 0.42 (0.36) is greater than 82% (93%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EIR strategy (red line).² Ignoring trading costs, a \$1 invested in the EIR strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides

would have yielded \$NaN which ranks the EIR strategy in the top 0% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EIR strategy would have yielded \$NaN which ranks the EIR strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EIR relative to those. Panel A shows that the EIR strategy gross alphas fall between the 62 and 74 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EIR strategy has a positive net generalized alpha for five out of the five factor models. In these cases EIR ranks between the 76 and 87 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does EIR add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations.

Figure 5 plots a name histogram of the correlations of EIR with 209 filtered anomaly of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EIR or at least to weaken the power EIR has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EIR conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EIR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EIR}EIR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EIR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EIR. Stocks are finally grouped into five EIR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EIR trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EIR and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EIR signal in these Fama-MacBeth regressions exceed

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

1.29, with the minimum t-statistic occurring when controlling for Net Payout Yield. Controlling for all six closely related anomalies, the t-statistic on EIR is 1.22.

Similarly, Table 5 reports results from spanning tests that regress returns to the EIR strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EIR strategy earns alphas that range from 19-29bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.41, which is achieved when controlling for Net Payout Yield. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EIR trading strategy achieves an alpha of 26bps/month with a t-statistic of 3.23.

7 Does EIR add relative to the whole zoo?

Finally, we can ask how much adding EIR to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the EIR signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes EIR grows to \$2832.91.

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EIR is available.

8 Conclusion

This study provides robust evidence that Equity Issuance Restraint (EIR) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that EIR generates substantial risk-adjusted returns that persist even after controlling for well-established factors and closely related equity financing signals.

The key findings reveal that a value-weighted long/short strategy based on EIR delivers impressive performance metrics, with an annualized net Sharpe ratio of 0.36 and statistically significant abnormal returns of 22 basis points per month relative to the Fama-French five-factor model plus momentum (t -statistic = 2.79). Most notably, the signal maintains its predictive power even when confronted with six closely related strategies from the factor zoo, including Net Payout Yield, Share Issuance measures, and Net Equity Financing, generating a monthly alpha of 26 basis points with a t -statistic of 3.23. This orthogonality to existing factors suggests that EIR captures unique information about firm quality and future performance not reflected in conventional equity issuance metrics.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For portfolio managers, the relatively modest degradation from gross to net returns (from 26 to 22 bps monthly) indicates that the strategy remains viable after accounting for realistic transaction costs. The signal's robustness across multiple factor models suggests it could serve as a valuable addition to multi-factor investment strategies, particularly for investors seeking exposure to firm financing decisions beyond traditional measures.

Several limitations of this study warrant consideration. First, while we employ comprehensive factor models, the possibility remains that EIR's predictive power may be explained by yet-unidentified risk factors or market frictions. Second, the

analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets remains an open question. Third, the economic mechanisms underlying EIR’s predictive ability deserve deeper investigation to distinguish between risk-based and behavioral explanations.

Future research should explore several promising avenues. Examining the performance of EIR across different market regimes and economic cycles would provide insights into its temporal stability. International evidence would help establish whether the signal’s effectiveness extends beyond U.S. markets. Additionally, investigating the interaction between EIR and corporate governance measures could illuminate whether managerial discipline plays a role in the signal’s predictive power. Finally, combining EIR with machine learning techniques might uncover non-linear relationships and enhance prediction accuracy. Understanding the precise economic channels through which equity issuance restraint affects future returns remains a critical area for theoretical development and empirical investigation.

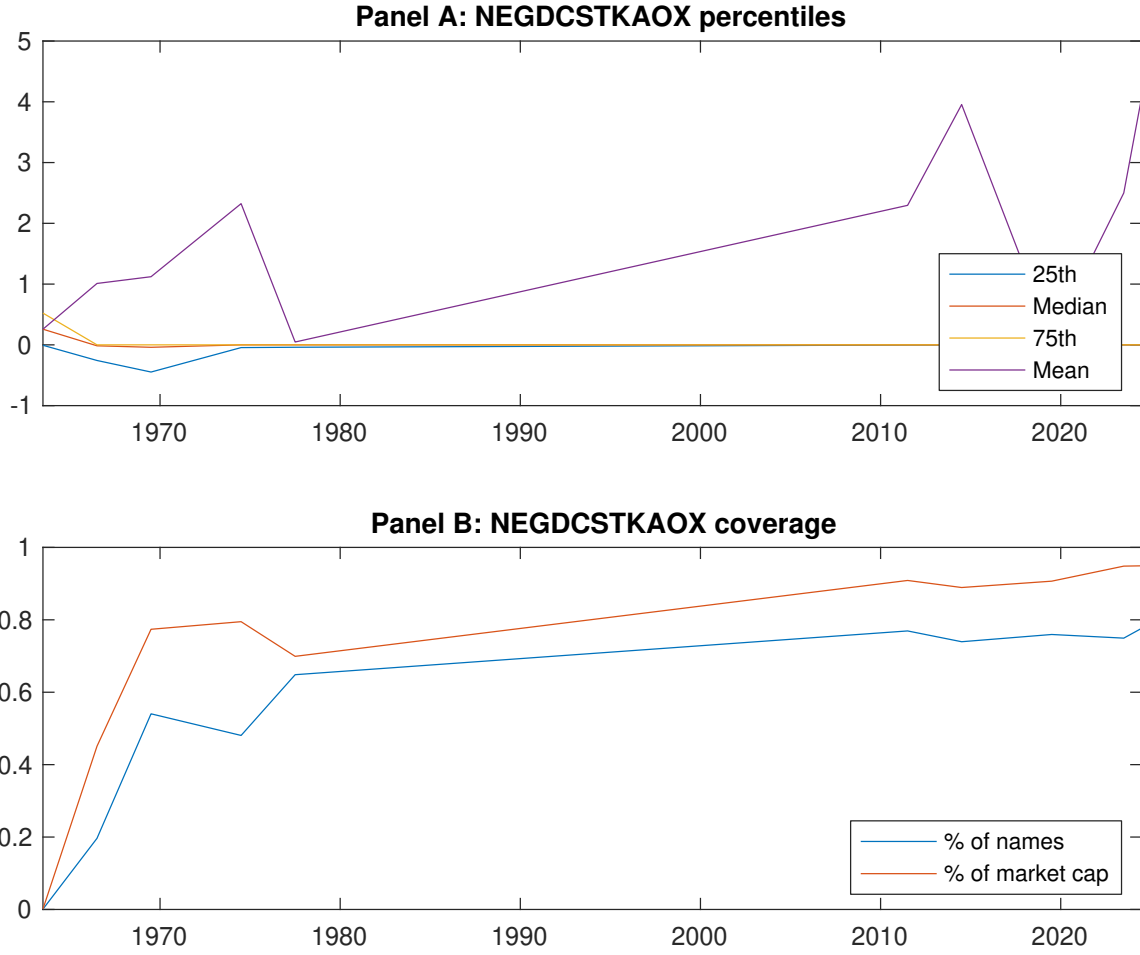


Figure 1: Times series of EIR percentiles and coverage.
This figure plots descriptive statistics for EIR. Panel A shows cross-sectional percentiles of EIR over the sample. Panel B plots the monthly coverage of EIR relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EIR. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on EIR-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.51 [2.74]	0.54 [2.89]	0.63 [3.43]	0.67 [3.99]	0.79 [4.85]	0.27 [3.25]
α_{CAPM}	-0.10 [-1.90]	-0.09 [-1.49]	0.02 [0.43]	0.12 [2.22]	0.24 [4.62]	0.34 [4.14]
α_{FF3}	-0.05 [-1.02]	-0.06 [-0.97]	0.05 [0.91]	0.08 [1.58]	0.22 [4.17]	0.27 [3.39]
α_{FF4}	-0.06 [-1.21]	-0.03 [-0.56]	0.06 [1.28]	0.04 [0.81]	0.22 [4.08]	0.28 [3.46]
α_{FF5}	-0.11 [-2.07]	-0.01 [-0.23]	0.06 [1.13]	-0.03 [-0.61]	0.14 [2.69]	0.24 [3.04]
α_{FF6}	-0.11 [-2.17]	0.00 [0.04]	0.07 [1.43]	-0.05 [-1.07]	0.15 [2.81]	0.26 [3.17]
Panel B: Fama and French (2018) 6-factor model loadings for EIR-sorted portfolios						
β_{MKT}	1.04 [82.63]	1.03 [69.83]	1.04 [84.13]	1.03 [85.65]	0.98 [77.13]	-0.06 [-3.29]
β_{SMB}	0.06 [3.45]	0.05 [2.39]	0.01 [0.66]	-0.07 [-4.03]	0.01 [0.59]	-0.05 [-1.90]
β_{HML}	-0.14 [-5.76]	-0.08 [-2.79]	-0.08 [-3.20]	0.06 [2.47]	-0.03 [-1.27]	0.11 [3.06]
β_{RMW}	0.17 [6.73]	-0.07 [-2.40]	-0.02 [-0.79]	0.17 [7.24]	0.11 [4.54]	-0.05 [-1.21]
β_{CMA}	-0.03 [-0.71]	-0.08 [-1.98]	-0.01 [-0.35]	0.24 [6.88]	0.21 [5.97]	0.22 [4.01]
β_{UMD}	0.01 [0.75]	-0.02 [-1.61]	-0.02 [-1.90]	0.03 [2.89]	-0.01 [-0.92]	-0.02 [-1.08]
Panel C: Average number of firms (n) and market capitalization (me)						
n	706	542	455	566	588	
me (\$10 ⁶)	1525	1339	1970	2150	2360	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EIR strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.27 [3.25]	0.34 [4.14]	0.27 [3.39]	0.28 [3.46]	0.24 [3.04]	0.26 [3.17]
Quintile	NYSE	EW	0.49 [6.32]	0.59 [8.02]	0.49 [7.63]	0.42 [6.59]	0.33 [5.49]	0.29 [4.76]
Quintile	Name	VW	0.30 [3.71]	0.35 [4.33]	0.29 [3.61]	0.29 [3.53]	0.28 [3.47]	0.28 [3.45]
Quintile	Cap	VW	0.20 [2.21]	0.27 [3.04]	0.21 [2.44]	0.21 [2.32]	0.19 [2.20]	0.19 [2.17]
Decile	NYSE	VW	0.25 [2.56]	0.30 [3.14]	0.21 [2.26]	0.20 [2.14]	0.24 [2.60]	0.24 [2.50]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.23 [2.78]	0.29 [3.60]	0.23 [2.89]	0.23 [2.95]	0.21 [2.64]	0.22 [2.79]
Quintile	NYSE	EW	0.29 [3.44]	0.40 [5.02]	0.30 [4.34]	0.27 [3.88]	0.15 [2.27]	0.14 [2.09]
Quintile	Name	VW	0.26 [3.22]	0.31 [3.78]	0.24 [3.11]	0.24 [3.08]	0.24 [3.01]	0.24 [3.07]
Quintile	Cap	VW	0.16 [1.80]	0.23 [2.59]	0.17 [2.00]	0.17 [1.95]	0.16 [1.92]	0.16 [1.92]
Decile	NYSE	VW	0.21 [2.13]	0.26 [2.62]	0.17 [1.85]	0.17 [1.79]	0.20 [2.12]	0.20 [2.16]

Table 3: Conditional sort on size and EIR

This table presents results for conditional double sorts on size and EIR. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EIR. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EIR and short stocks with low EIR. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	EIR Quintiles					EIR Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.44 [1.60]	0.74 [2.77]	0.86 [3.21]	0.94 [3.72]	1.11 [4.56]	0.66 [5.72]	0.74 [6.44]	0.64 [6.02]	0.61 [5.66]	0.46 [4.39]	0.45 [4.24]
	(2)	0.55 [2.21]	0.74 [2.98]	0.79 [3.28]	1.01 [4.27]	1.00 [4.41]	0.43 [4.15]	0.51 [5.09]	0.39 [4.21]	0.36 [3.80]	0.32 [3.40]	0.30 [3.14]
	(3)	0.67 [2.92]	0.67 [2.87]	0.84 [3.67]	0.94 [4.27]	0.93 [4.52]	0.26 [2.89]	0.34 [3.79]	0.24 [2.86]	0.23 [2.66]	0.18 [2.08]	0.17 [1.99]
	(4)	0.49 [2.25]	0.68 [3.15]	0.85 [4.02]	0.88 [4.36]	0.89 [4.63]	0.41 [4.18]	0.49 [5.15]	0.36 [4.34]	0.37 [4.38]	0.22 [2.67]	0.24 [2.90]
	(5)	0.55 [3.09]	0.49 [2.68]	0.44 [2.44]	0.57 [3.36]	0.77 [4.77]	0.20 [2.19]	0.25 [2.69]	0.20 [2.16]	0.21 [2.25]	0.24 [2.55]	0.25 [2.63]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	EIR Quintiles					EIR Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	315	314	314	313	313	25	28	32	24	24	
	(2)	91	90	90	90	89	48	49	50	49	48	
	(3)	65	65	65	64	65	85	85	86	88	88	
	(4)	55	55	55	55	55	185	187	196	194	199	
(5)	51	51	51	51	50	1352	1347	1644	1482	1780		

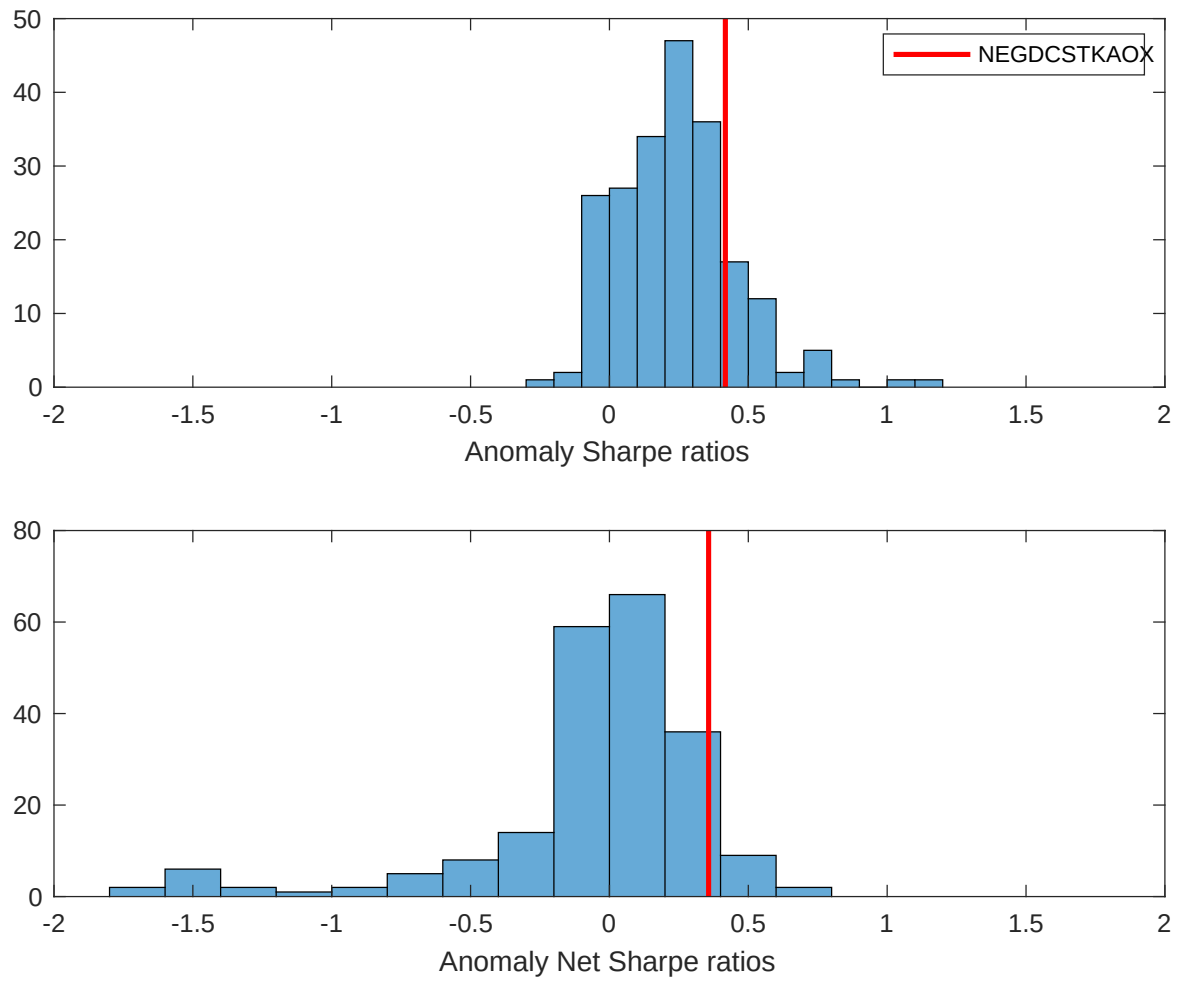


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EIR with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

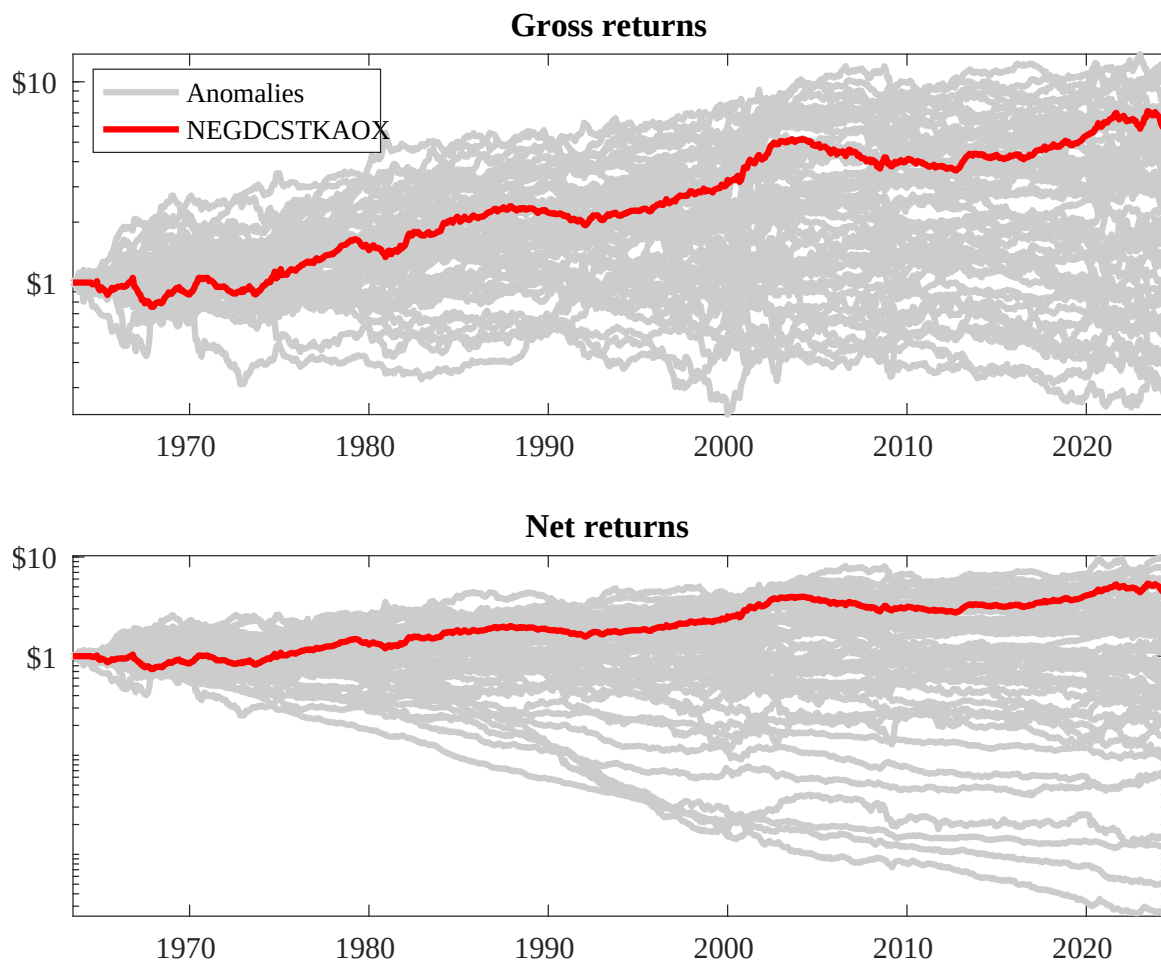
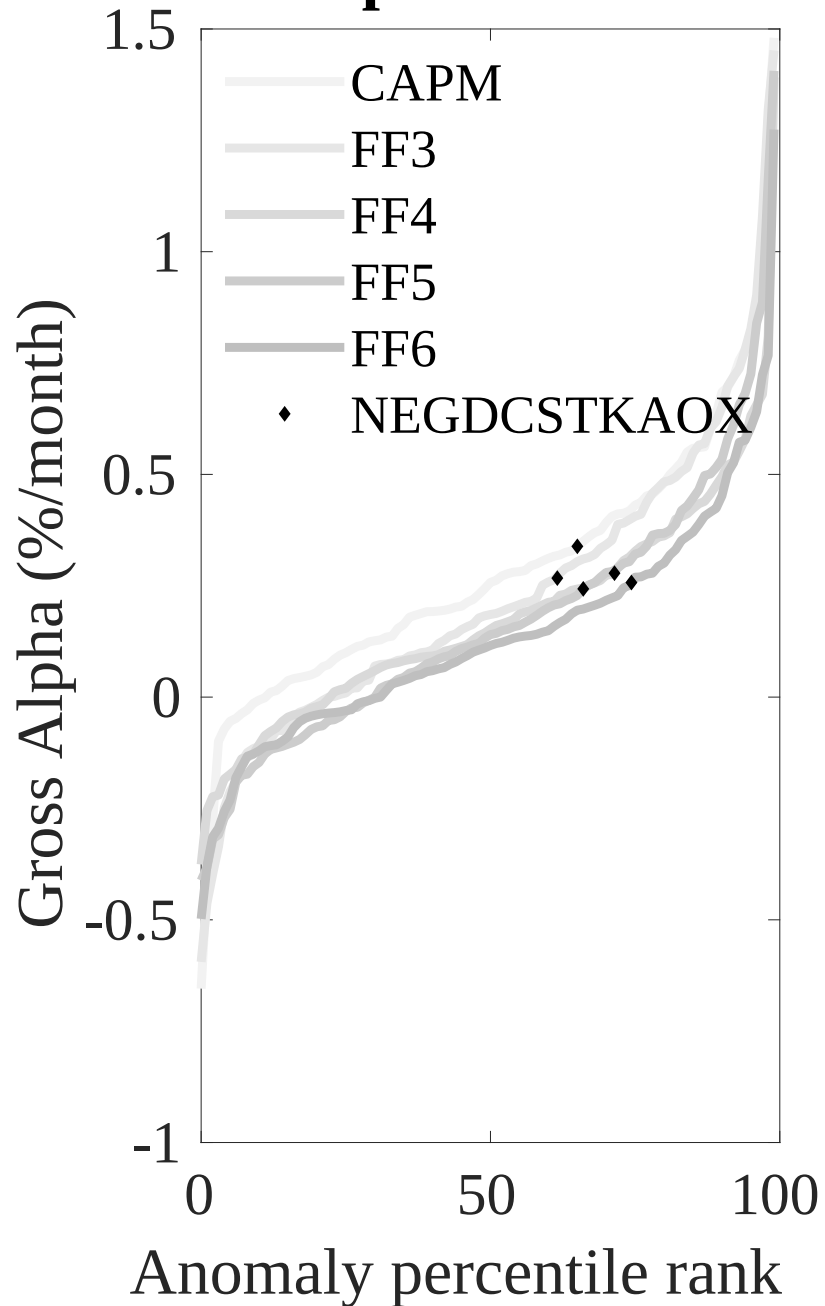


Figure 3: Dollar invested.

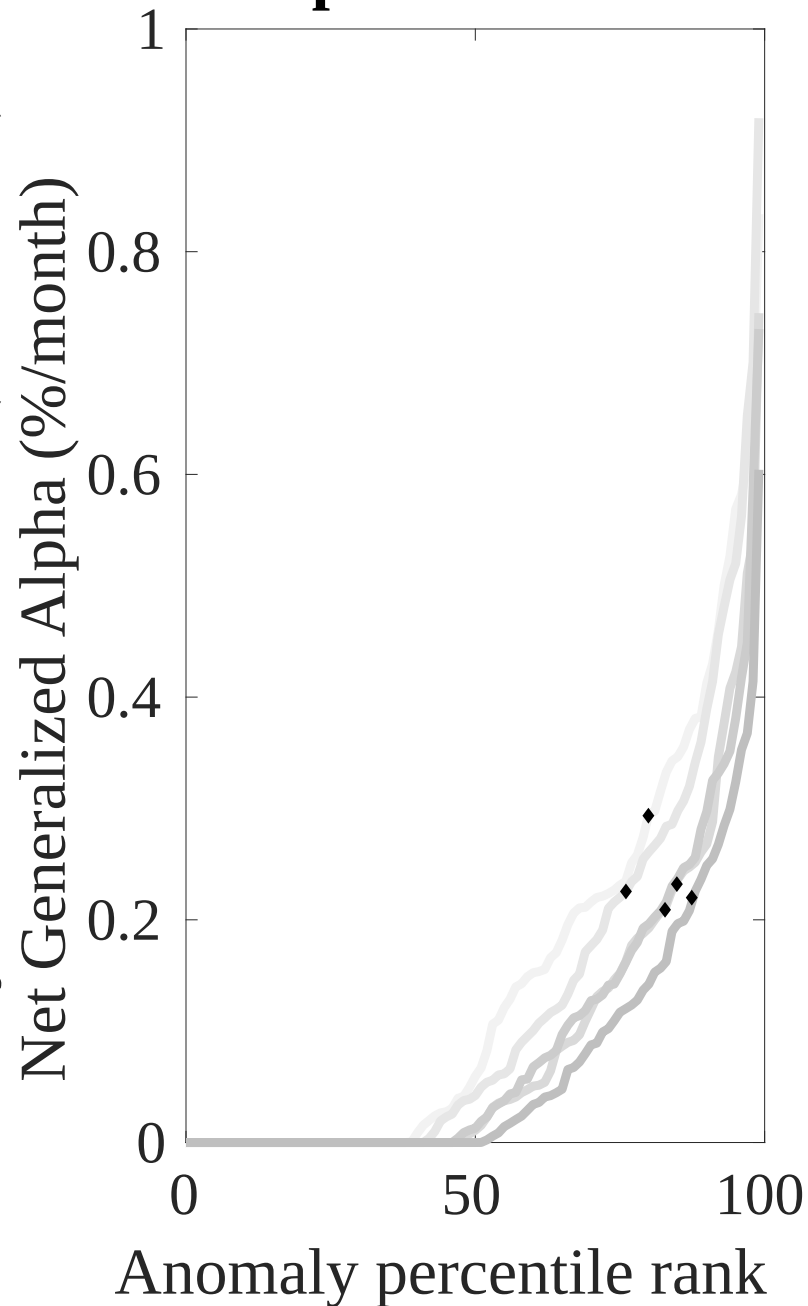
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EIR trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



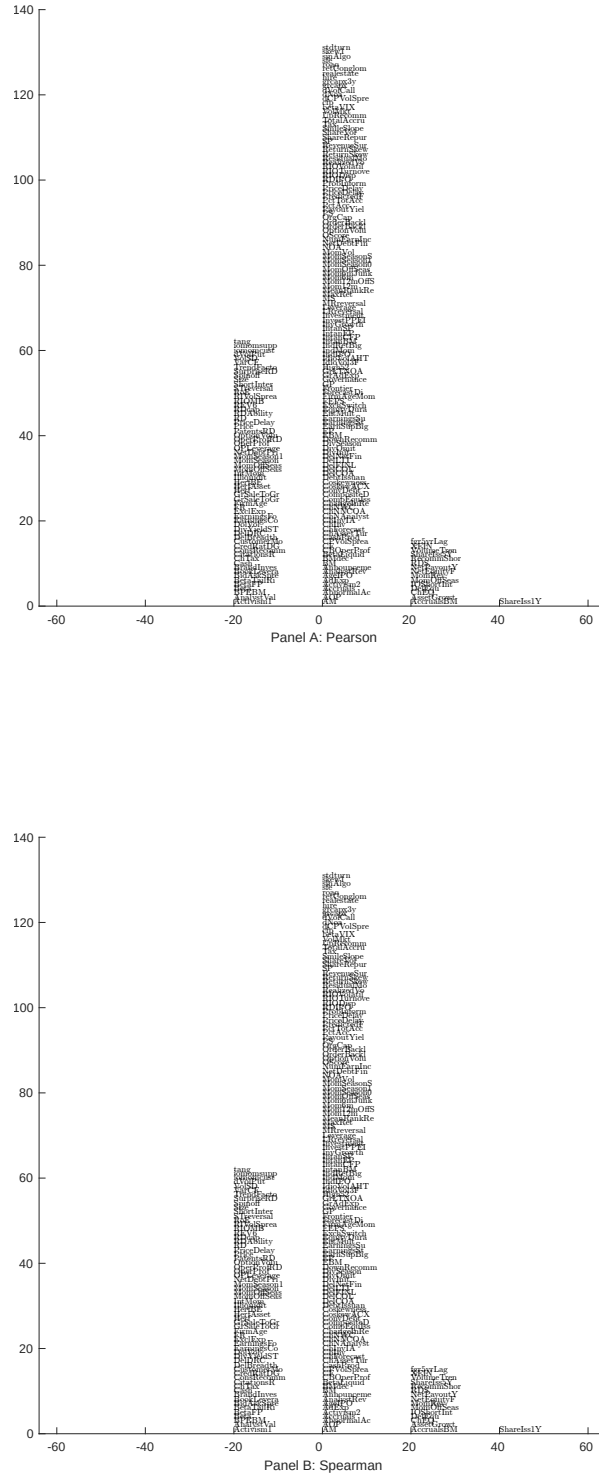


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 209 filtered anomaly signals with EIR. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

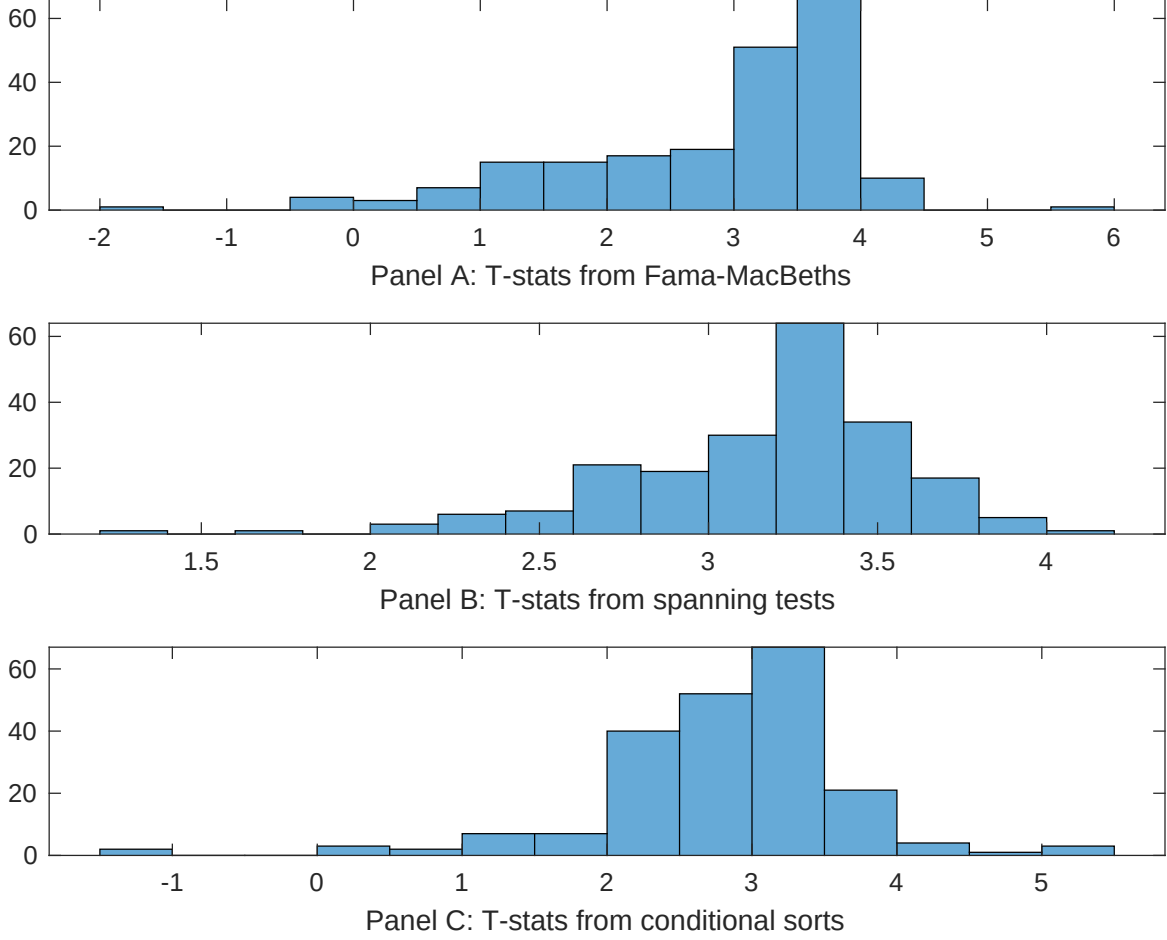


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of EIR conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EIR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EIR}EIR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EIR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EIR. Stocks are finally grouped into five EIR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EIR trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EIR. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EIR}EIR_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Net Payout Yield, Growth in book equity, Share issuance (1 year), Change in equity to assets, Share issuance (5 year), Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.12 [5.34]	0.18 [7.25]	0.13 [5.84]	0.13 [5.54]	0.14 [6.23]	0.13 [5.31]	0.12 [4.61]
EIR	0.30 [1.29]	0.46 [2.04]	0.57 [2.72]	0.63 [2.71]	0.56 [2.81]	0.53 [2.21]	0.30 [1.22]
Anomaly 1	0.24 [2.30]						0.14 [2.78]
Anomaly 2		0.52 [5.12]					-0.16 [-1.12]
Anomaly 3			0.16 [6.93]				0.90 [3.85]
Anomaly 4				0.16 [4.76]			0.13 [2.18]
Anomaly 5					0.29 [5.40]		0.28 [6.46]
Anomaly 6						0.15 [2.40]	-0.15 [-1.83]
# months	703	708	703	708	703	630	622
$\bar{R}^2(\%)$	1	0	0	0	0	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EIR trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EIR} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Net Payout Yield, Growth in book equity, Share issuance (1 year), Change in equity to assets, Share issuance (5 year), Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.28 [3.60]	0.23 [3.01]	0.24 [3.04]	0.25 [3.15]	0.19 [2.41]	0.29 [3.47]	0.26 [3.23]
Anomaly 1	20.23 [6.65]						17.33 [4.24]
Anomaly 2		32.24 [7.59]					32.13 [4.93]
Anomaly 3			17.48 [4.32]				-9.66 [-1.81]
Anomaly 4				17.24 [4.32]			-6.74 [-1.12]
Anomaly 5					23.63 [5.57]		17.10 [3.23]
Anomaly 6						12.43 [3.24]	-6.78 [-1.41]
mkt	-3.12 [-1.63]	-5.19 [-2.81]	-5.75 [-3.03]	-6.55 [-3.45]	-5.13 [-2.72]	-4.79 [-2.30]	-3.32 [-1.63]
smb	-0.92 [-0.34]	-5.26 [-1.97]	-3.66 [-1.33]	-4.40 [-1.61]	-2.73 [-1.00]	3.09 [0.93]	-1.79 [-0.55]
hml	4.55 [1.19]	8.64 [2.39]	12.26 [3.34]	10.31 [2.78]	14.54 [3.99]	11.48 [3.02]	5.38 [1.39]
rmw	-16.52 [-4.07]	-2.90 [-0.80]	-14.08 [-3.27]	-2.90 [-0.78]	-13.13 [-3.30]	-8.43 [-1.87]	-7.88 [-1.61]
cma	4.49 [0.77]	-10.20 [-1.52]	10.38 [1.75]	3.82 [0.57]	9.87 [1.73]	7.41 [1.20]	-21.25 [-2.93]
umd	-0.36 [-0.19]	-2.46 [-1.34]	-2.70 [-1.43]	-1.48 [-0.79]	-3.45 [-1.83]	-1.98 [-1.01]	-1.92 [-1.01]
# months	716	720	716	720	716	630	626
$\bar{R}^2(\%)$	21	22	18	18	19	14	23

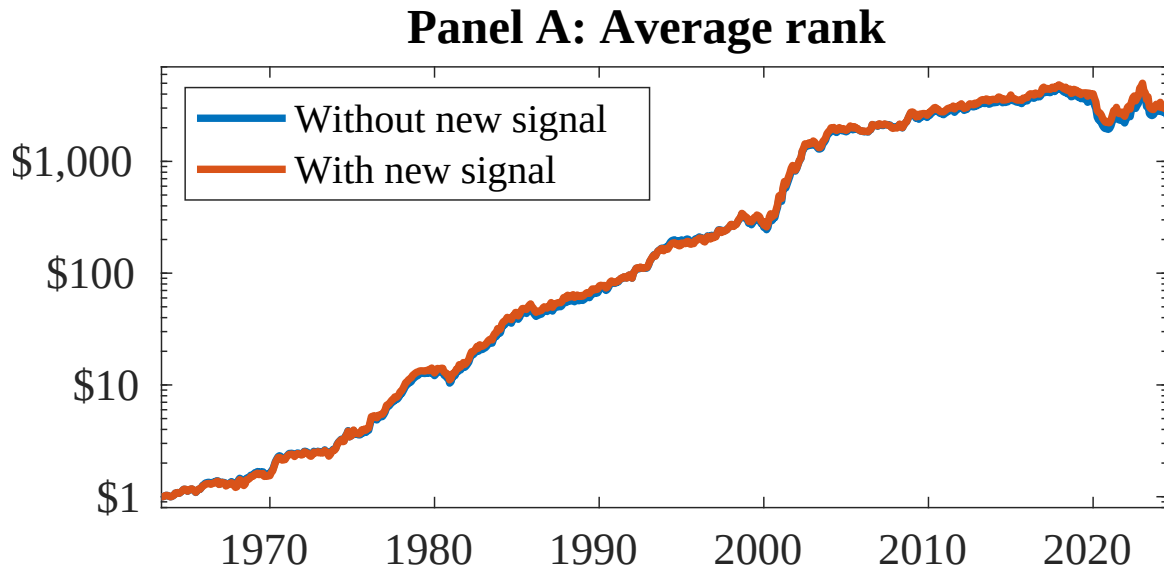


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as EIR. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

References

- Bansal, R. and Yaron, A. (2004). Risks for the long run: A potential resolution of asset pricing puzzles. *Journal of Finance*, 59(4):1481–1509.
- Barro, R. J. (2006). Rare disasters and asset markets in the twentieth century. *Quarterly Journal of Economics*, 121(3):823–866.
- Campbell, J. Y. and Cochrane, J. H. (1999). By force of habit: A consumption-based explanation of aggregate stock market behavior. *Journal of Political Economy*, 107(2):205–251.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.
- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.
- Constantinides, G. M. (1986). Capital market equilibrium with transaction costs. *Journal of Political Economy*, 94(4):842–862.
- Daniel, K. and Titman, S. (2006). Market reactions to tangible and intangible information. *Journal of Finance*, 61(4):1605–1643.
- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance*, Forthcoming.
- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.

- Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.
- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.
- Hansen, L. P. and Scheinkman, J. A. (2009). Long-term risk: An operator approach. *Econometrica*, 77(1):177–234.
- Lyandres, E. and Michenaud, S. (2023). Seasoned equity offerings and corporate financial flexibility. *Journal of Financial Economics*, 148(2):186–206.
- McLean, R. D. and Pontiff, J. (2016). Does academic research destroy stock return predictability? *Journal of Finance*, 71(1):5–32.
- Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper*.
- Pontiff, J. and Woodgate, A. (2008). Share issuance and cross-sectional returns. *Journal of Finance*, 63(2):921–945.
- Zhang, L. (2005). The value premium. *Journal of Finance*, 60(1):67–103.